



BI Solution for Bean Counter Cafe

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Introduction to Bean Counter Cafe

In 1988, Alice Lombardi started Bean Counter Bakery Café out of her licensed home kitchen in Worcester. After years of community support, she was able to open her first café on Highland Street in Worcester, Massachusetts in 2001. Then expanding to another location on Grove Street in May of 2020. Over the last 20 years, Bean Counter Café was able to grow from a one-person operation, to a multi location bakery business in Worcester, Massachusetts.



Intro Contd & Client Company Background

Bean Counter Bakery Café focuses on local bakery and café needs, as well as the creation of specialty coffees, baked goods, and breakfast/lunch items. They are a local company with two locations surrounding WPI. Their store includes the store management, the front-of-house staff, and the kitchen and bakery staff.



Bean Counter Café issues

However, despite years of loyalty to the surrounding community, it took 13 years to save up enough to afford a location. Our business intelligence team was tasked with analyzing Dunkin Donut's Stores datasets to see how insights in similar large café's can be applied to Bean Counter Bakery Café. Our team analyzed sales, location, revenue, and foot traffic by different stores, as well as two interactive dashboards and implementation recommendations for Bean Counter.



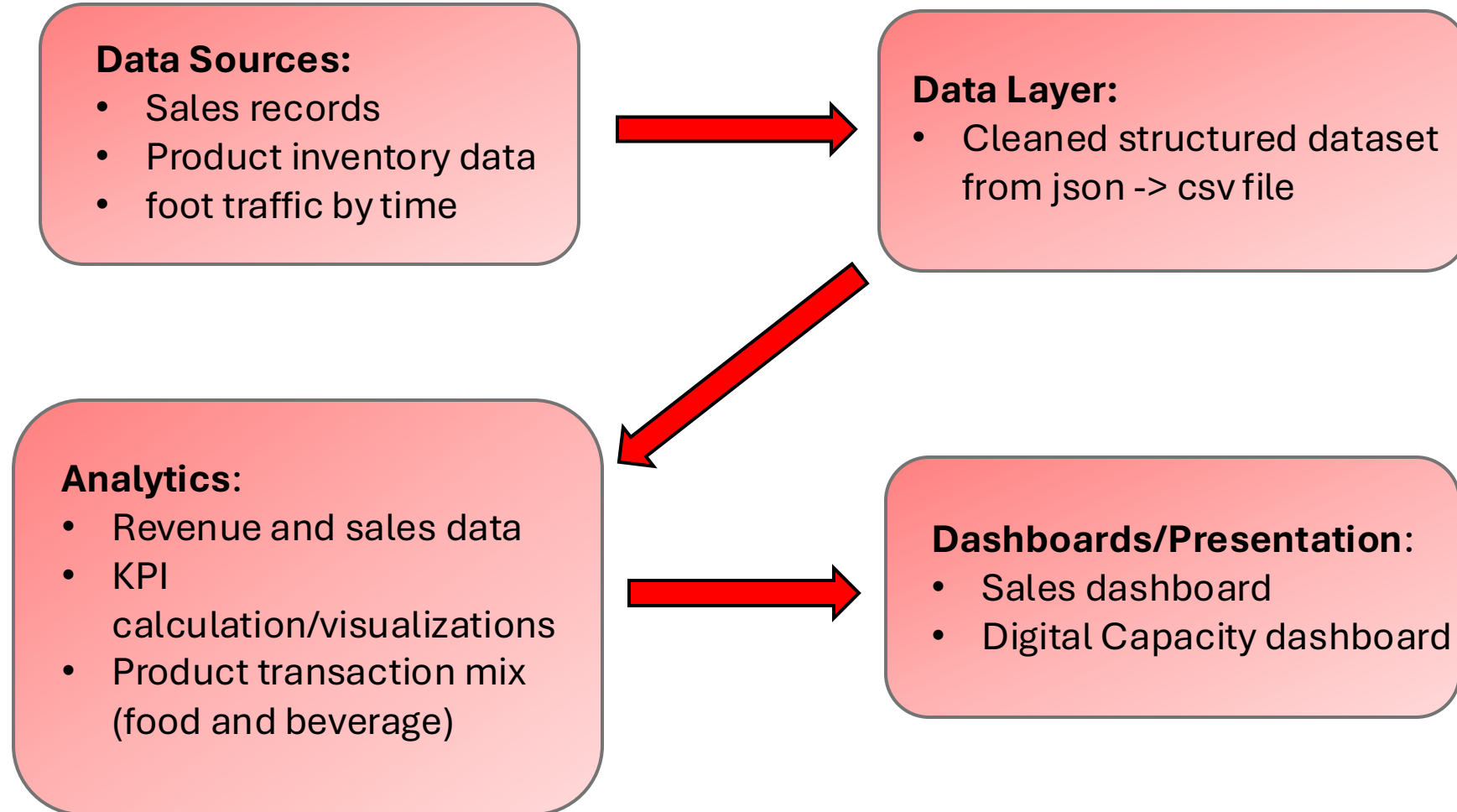
Expected value of BI for client company

What is BI?

Business intelligence (BI) is the broad term for the technologies, applications, and processes used for gathering, cleaning, storing, analyzing, and visualizing data to help users make better decisions.

BI can help businesses with problems involving data management, data reporting, performance and operational inefficiencies, sales challenges, and risk management.

BI Framework for Bean Counter



Case Study: McDonalds

Using ERP and BI deployment,
McDonalds was able to

- Expose data quality issues
- Fill significant performance, data quality, and scalability deficiencies
- Resolve performance issues that had been impacting user adoption



Dashboard One: Sales Dashboard

Intended users: Sales executives and managers

Managerial Questions:

- What time of day does the café make the most sales?
- What drinks are most profitable?
- What donuts are most profitable?
- What location makes the most sales?
- What month does the café make the highest sales?

Dashboard Two: Digital Capacity Dashboard

Intended users: Store owners and executives

Managerial Questions:

- How digitally enabled are our stores?
- Where should we be looking to expand to?
- Are we investing in the right markets?
- Which locations lack digital capabilities?

Discussion: Managerial Aspects

Bean Counter should choose performance indicators that align with their strategic goals such as increasing sales, improving customer satisfaction, and expansion as shown in our proposed BI solutions.

Potential KPIs:

- Daily revenue
- Sales by product category
- Peak-hours sales trends
- Customer return rate
- Customer satisfaction
- Mobile app usage by location
- Average foot traffic by location

Discussion: Technical Aspects

Bean Counter relies on multiple data sources such as:

- Sales records
- Inventory records
- Supplier invoices
- Online ordering platforms.

Could have issues involving:

- Inconsistency with product naming
- Duplicate records
- Inaccurate inventory counts
- Delays with updating information

To combat this, we:

- Use software, such as Tableau, to help clean and validate data
 - Helps flag unusual entries (with potentially incorrect data) or duplicate information.
- Integrate databases to reduce discrepancies within the data

Discussion: Ethical and Legal Aspects

If Bean Counter Café opens in Europe, it must comply with the General Data Protection Regulation (GDPR). Article 44 requires that any transfer of personal data outside the EU maintains equivalent data protection standards. Article 21 gives customer the right to object to their personal data being used for direct marketing, including promotional communications.

Some solutions to mitigate these articles include:

- Implementing a data protection officer, as this person can be the one to mediate between the companies' regulatory authorities, and relevant data subjects.
- Implementing a privacy policy for people trying to understand how data is being stored and collected
 - Make sure the privacy policy is read and understood

Other Processes and Artifacts

Social Media Analytics can help us:

- Identify what processes work for the company
- Discover customer thoughts on different operations/food
- Determine what processes can be improved, which can optimize our dashboards

Some important techniques that can be useful are

- Sentiment Analysis
- Social Network Analysis

Summary and Conclusion

In the Sales dashboard, some insights to summarize the findings include

1. Total sales was higher for the donut than drink sales.
2. We also noticed a combination of higher sales in winter months and during morning hours.
3. The two locations with the highest sales are in LA and NY.

In the Digital Capacity dashboard, we noticed there was

1. A lot of locations have mobile ordering and digital platforms
 - Above 65 % of locations have at least 1 digital platform
2. There are less locations with pop-up kiosks in drive-thru

We recommend locations to invest in more technology, and to innovate in that sector. The data supports the stores with digital innovation, as NY regions also have high sales and at least 1 digital feature.

References

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